Al Imran Alif

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PROFILE

Computer Science student and full-stack developer with a software engineering focus and a strong foundation in data structures, algorithms, object-oriented programming, and software design patterns. Solved algorithmic problems through competitive programming, strengthening analytical thinking and problem-solving skills. Build modular web applications and REST APIs using Node.js, NestJS, Python, and FastAPI, with growing experience in Django. Develop responsive interfaces with React, Next.js, and Vue.js, supported by MySQL, Oracle, PostgreSQL, and MongoDB. Also a Student Researcher at ELITE Research Lab LLC, exploring machine learning, deep learning, medical AI, bio-sustainability, and bioinformatics.

EDUCATION

American International University-Bangladesh — **Dhaka, Bangladesh**
B.Sc. in Computer Science and Engineering — *Jan 2023 – Dec 2026 (Expected)*

EXPERIENCE

ELITE Research Lab LLC
Student Researcher — *Aug 2026 -Current*
Research focus: machine learning, deep learning, medical AI, bio-sustainability, and bioinformatics.

DEVELOPMENT PROJECTS

PowerPlan
Load-Shedding Window Planner
Live Demo | GitHub

- Built and deployed a full-stack scheduling application that places business jobs around multiple daily power-cut windows while respecting three electricity-requirement modes.
- Automated generator-minute and operating-cost calculations, unscheduled-job detection, schedule recalculation, and visualization on a 24-hour timeline.
- Persisted outage windows and jobs in PostgreSQL and validated REST API requests with Zod across the deployed frontend and backend.

Tools: Next.js, React, TypeScript, Node.js, Express, PostgreSQL, Zod, Vercel, Render, Neon

ReconDesk
Explainable Two-Source Sales Reconciliation Engine
Live Demo | GitHub

- Developed and deployed a reconciliation engine for two inconsistent datasets containing 90 and 93 transactions, including missing, duplicate, fee-adjusted, and time-shifted records.
- Implemented explainable matching with weighted reference, amount, and time signals, plus confidence thresholds, human review, manual pairing, and immediate dashboard recalculation.
- Persisted accepted, rejected, and manually paired decisions in PostgreSQL so review outcomes remain consistent after refresh.

Tools: Next.js, React, TypeScript, Node.js, Express, PostgreSQL, Zod, Vercel, Render, Neon

Swap Owner
Role-Based Buy and Sell Marketplace
GitHub

- Developed a full-stack marketplace with customer and manager authentication, product browsing, product details, carts, orders, reviews, profiles, and seller management workflows.
- Implemented token-authenticated API routes for profile updates, seller listings, shopping carts, order placement, and administrative views.
- Organized customer, manager, and marketplace functions into separate backend modules and responsive Next.js routes for maintainable development.

Tools: Next.js, React, TypeScript, NestJS, PostgreSQL, TypeORM, JWT, Axios, Tailwind CSS

RESEARCH PROJECT

Cold-Chain Temperature Monitoring
Event-Driven Monitoring and Tamper-Detection Experiment
GitHub

- Generated 1,000 synthetic temperature readings across 20 simulated vaccine batches and four shipment stages using fixed random seeds.
- Evaluated threshold-based excursion detection under simulated sensor noise; reproduced 96.5% accuracy, 98.65% recall, and 0.893 F1 on the synthetic dataset.
- Simulated alerts and quarantine responses; detected all altered records in separate 10-, 25-, and 50-record SHA-256 tampering tests and exported results to CSV.

Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib, SHA-256

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C, C++, C#, PHP,JavaScript (ES6+), TypeScript,.
Backend Development: Node.js, NestJS, Express.js, FastAPI, REST API design, MVC, middleware, JWT authentication, RBAC, bcrypt, asynchronous programming; Django (learning).
Frontend Development: React, Next.js, Vue.js, HTML5, CSS3, Tailwind CSS, responsive design.
Databases: MySQL, Oracle Database, PostgreSQL, MongoDB, SQL Server, TypeORM, normalization, schema design, query optimization.
Testing & QA: Selenium, browser automation, functional testing, Postman API testing, debugging, request validation.
Tools & Workflows: Git, GitHub, Swagger, Postman, VS Code, Figma, basic Linux, Agile/sprint-based development, Vercel, Render, Neon , superbase.
AI-Assisted Development: ChatGPT, Claude, Cursor, Codex for coding support, debugging, testing, research, and documentation.
Core Concepts: Data structures & algorithms, OOP, database design, software engineering, software architecture and design patterns, system design.
Research Areas: Machine learning, deep learning, medical AI, bioinformatics, computational biology, and AI for bio-sustainability.

CERTIFICATIONS

CS Fundamentals – Batch 7 **Phitron**
C, C++, Data Structures & Algorithms *Mar – Sep 2025*

- Completed training in programming fundamentals, data structures, and algorithms through structured problem-solving practice.

AI Essentials – Grameenphone Academy
AI Workflows & Agents; Generative AI & Prompt Engineering; Online Safety Fundamentals; Personal Branding in the AI Era.

TRAINING & ACTIVITIES

- Solved 400+ algorithmic problems across LeetCode, Codeforces, and HackerRank, applying data structures, algorithms, and debugging skills.
- Practice competitive programming on CodeChef and participate in online coding challenges.
- Participated in AIUB CS Fest as a competitive programmer.
- Strengthened OOP, debugging, database design, and software engineering skills through hands-on problem solving and learning unfamiliar technologies.
- Applied software engineering principles and architectural design patterns to structure modular, maintainable applications with clear separation of concerns and support for future growth.